

DATA ANALYTICS · INTERMEDIATE

Data Analytics Path

Turn raw data into a clear decision.



Turn raw data into trustworthy analysis and decision-ready stories.

🕒 10-16 weeks

📁 4 modules

📖 16 lessons

[Explore the course →](#)

Concepts, worked examples, assignments and a coherent course project.

WHO IT IS FOR · WHAT YOU WILL LEARN

Your next capability.

A hands-on path through Excel, SQL, Python, practical statistics, visualization, BI dashboards and stakeholder communication.



Query and prepare data



Build trustworthy dashboards



Apply practical statistics



Communicate decision-ready insights



Audience

- Analysts
- Developers moving into data
- Business professionals



Prerequisites

- Basic spreadsheet familiarity

FROM CONCEPT TO EVIDENCE



One curriculum, three connected views

This guide summarizes the same modules and lessons shown on the course overview and in the learning workspace. The next pages explain the scope and practical evidence for every module.

DETAILED MODULE CURRICULUM · 1-2

Build the foundations.

The module names, lesson sequence and evidence below match the course learning workspace.

1 Analytics Foundations

Turn a stakeholder question into a metric that can be reproduced.

1.1 Business questions and KPIs

Define the decision before selecting a chart.

1.2 Excel analysis

A spreadsheet analysis should preserve raw data and separate calculations from presentation.

1.3 Data quality

Data quality includes completeness, validity, uniqueness and consistency.

1.4 Analytical thinking

Analytical thinking distinguishes observation, interpretation and recommendation.

You will produce: Metric dictionary and reconciled analysis.

2 SQL and Data Preparation

Build an analysis dataset with a known row grain.

2.1 SQL foundations

Start a SQL query by identifying what one output row represents.

2.2 Joins and aggregation

Joins can multiply rows when keys are not unique.

2.3 Cleaning workflows

A cleaning workflow should be repeatable and preserve an audit trail.

2.4 Reusable analysis datasets

A reusable dataset includes its grain, keys, definitions and refresh assumptions.

You will produce: SQL queries, cardinality checks and dataset contract.

Practice is part of every module

Work through the example, record your reasoning, complete the module assignment and compare your work with the review criteria.

DETAILED MODULE CURRICULUM · 3-4

Apply, integrate and demonstrate.

The module names, lesson sequence and evidence below match the course learning workspace.

3 Python and Statistics

Explore a dataset without confusing patterns with proof.

3.1 Pandas workflows

A Pandas workflow should make selection, transformation and aggregation readable.

3.2 Exploratory analysis

Exploratory analysis checks distributions, outliers and relationships before a final narrative.

3.3 Practical statistics

Choose summary statistics that fit the distribution and question.

3.4 Experiment interpretation

Experiment interpretation needs a design, outcome definition and appropriate uncertainty assessment.

You will produce: Notebook, exploratory plots and interpretation memo.

4 BI and Storytelling

Communicate a decision with readable visuals and honest caveats.

4.1 Visualization principles

Choose visual encodings that fit the comparison.

4.2 Dashboard design

A dashboard should support a small set of user decisions.

4.3 Stakeholder narrative

A stakeholder narrative connects the question, method, finding, limitations and proposed action.

4.4 Analytics capstone

The capstone combines a reproducible dataset, analysis, dashboard and recommendation.

You will produce: Dashboard, source queries and stakeholder memo.

Practice is part of every module

Work through the example, record your reasoning, complete the module assignment and compare your work with the review criteria.

PROJECTS AND WORKING METHODS

Decision-ready analytics report

Investigate a practical business question, prepare the data, explore patterns and communicate findings with a clear dashboard and caveats.



Documented queries and cleaning



Interactive dashboard



Evidence-based recommendation



Tools and methods

Excel

SQL

Python and Pandas

Power BI or Tableau

Tools are learning choices, not partner endorsements.

Inside module 1: a worked example

Cohort A: 80 completions / 100 eligible = 80%

Cohort B: 60 completions / 60 eligible = 100%

A has more completions; B has the higher completion rate

Check equal follow-up periods before comparing.

A missing denominator can reverse the interpretation without changing any completion count.

How your work is reviewed

Show the artifact, explain your decisions, test the stated assumptions and identify limitations. Module assignments build toward the course project rather than replacing it with a single example.

DELIVERY, COMPLETION AND NEXT STEPS

Know what you are signing up for.

**Delivery**

Self-paced online · 10–16 weeks

Confirm current schedule, cohort arrangements and enrollment terms with the academy.

**Completion requirements**

Complete the required coursework and module assignments, and pass the course assessment. A course completion certificate is available after those requirements are met.

What the learning workspace includes**Follow the modules**

Expand the curriculum and open the exact lesson you need.

**Apply the concepts**

Read explanations, inspect examples and complete module practice.

**Keep your evidence**

Use resources, notes, discussion and assignments to support your learning.

Learning commitments, not outcome guarantees

Completion is not a placement, employment or performance guarantee. The course project is educational evidence; any production use requires appropriate review and validation.

Start with the curriculum. Continue with the first lesson.

Review your starting point, download this guide and explore the connected course experience.

[Go to the first lesson](#) →

Client-review edition. The static learning workspace saves demonstration progress locally; production enrollment, assessment, communications and certificate issuance are not connected.